

27 B

$$L = \frac{N}{t} \frac{hf}{A}$$

If the frequency is increased to $2f$, the no. of photons incident per unit time (N/t) will be halved for the same intensity L and hence the number of photoelectrons emitted per second will decrease. Hence, the maximum photocurrent will be less than I .

Using the Einstein photoelectric equation, $hf = eV_s + \phi$, if the frequency is increased to $2f$, the photoelectric equation will be $h2f = eV'_s + \phi$, hence $eV'_s + \phi = 2eV_s + 2\phi$

$$\Rightarrow V'_s = 2V_s + \frac{\phi}{e}$$

28 B

Frequency of uv > frequency of yellow > frequency of red.